Can disease spread be modelled by a reaction-diffusion equation?

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Abstract

In this study, we modelled the spread of disease in Manchester through a reaction-diffusion equation, comparing this to pre-lockdown Covid data. The main assumptions of the model were a constant population density, a recovery time of 14 days, and effects on immunocompromised people is ignored. The Euler method is used to solve the equation computationally. Our study successfully/unsuccessfully modelled spatial disease spread when comparing to pre-pandemic Covid data, but only the initial spread. The model could be improved by implementing random-walks to model more random behaviour, modelling more outbreak centres, or accounting for disease spread reduction.

Introduction

Modelling disease spread is important for governments have appropriate plans in case of large-scale pandemics. The recent Covid pandemic was not handled very well (give some examples, probably cite some things here). With this motivation, we attempted to model the disease spread within greater Manchester. Using our knowledge of physics, we supposed that the spread of a disease like Covid could be modelled by a diffusion equation.

However, due to mass conservation, a standard diffusion equation is incapable of accurately modelling disease spread, because diseases are inherently create new infections. Therefore, for disease modelling, a reaction-diffusion equation is much so that the number of infections at any time is not conserved. This allows us to model new infections and patient recovery.

Theory - Reaction-Diffusion Equation

We attempt to model the spread of disease via a reaction-diffusion equation,

$$\frac{\partial I}{\partial t} = D\nabla^2 I + \beta(x,y)SI - \gamma I, \tag{1}$$

which models spatial densities ρ, which are parametrised such that,

$$\rho = S + I + R, \tag{2}$$

where,

- $S(x,y,t)$: Susceptible population.
  - Those who can catch disease.
- $I(x,y,t)$: Infected population.
  - Those who have the disease.
- $R(x,y,t)$: Recovered population.
  - Those who are no longer infected.

Results: Simulation vs Data

Infection of COVID-19 was modelled in Greater Manchester with an initial transmissibility of $\beta_0 = 0.48$ per day taking into account weather effects [1] while using a diffusion coefficient of $D = 0.85$ m^2 s^{-1}, and a mean recovery time of $\tau = 7$ days [2] and using the spatial density given for each county of Greater Manchester from the March 2021 Census data [3]. The model was compared to the COVID-19 infection map of Boston [4] as it is a city of comparable size thus allowing us to clearly compare the accuracy of our model. As seen our model is comparable to the Boston infection map up to the first 30 days. After that, the model diverges and while spreading out the infection at the initial points in the centre of Greater Manchester disappears as opposed to the Boston heat map.

Results: Model Figure

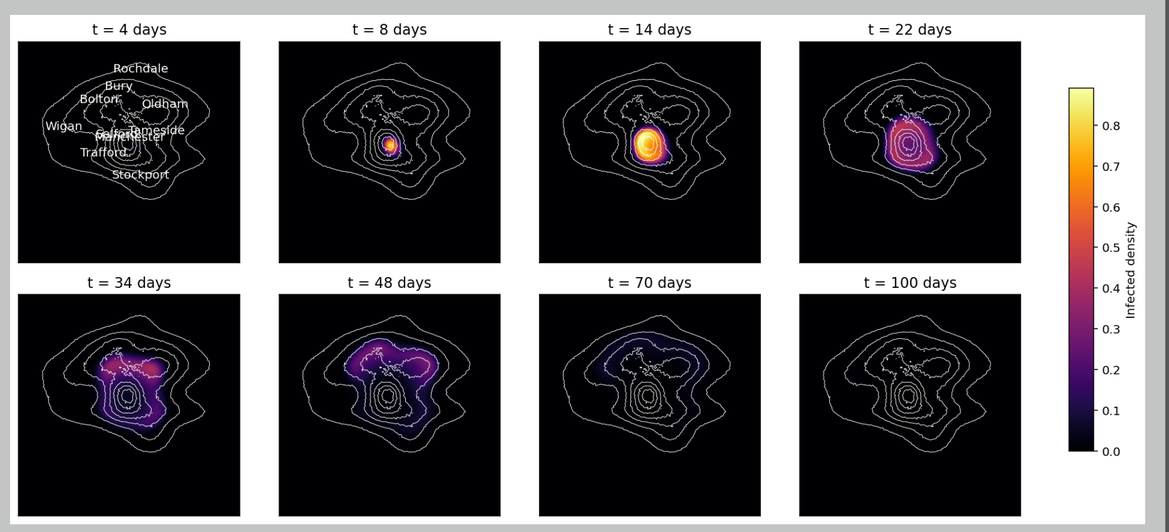

Figure 1: COVID-19 infection over time in Greater Manchester up to 100 days. Using parameters of initial transmission rate $\beta_0 = 0.48 perday$ and diffusion coefficient of $D = 0.85 m^2 s^{-1}$ and a mean recovery time of $\tau = 7 days$.

Results: Simulation vs Data

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Results: Spread of Covid From Data

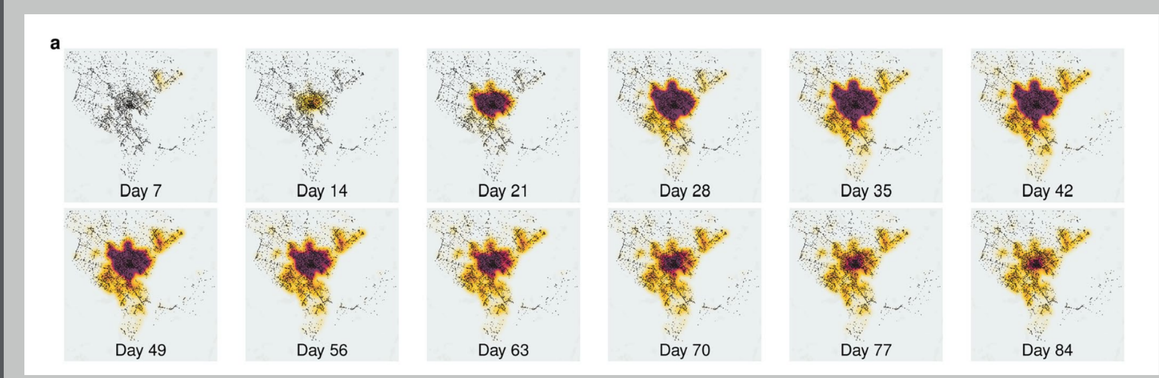

Figure 2: Covid-19 infection "Heatmap of infected (exposed and infectious) individuals every seven days: (a) Auto Sprawl;"[4] of Boston over 84 days.

Discussion

We found that though our model was successful at modelling the initial spread of the Covid 19 pandemic. In particular, the initial growth seen in Fig.1 matches near perfectly data seen in Fig. 2. However, after this, the model starts to diverge from real life data. This is likely because we made the assumption that people's behaviour wouldn't change when infected and the nature of the Reaction-Diffusion equation used in the model. In reality, as time went on, and more was understood about Covid, people started to self isolate when they first experienced symptoms, and the first lockdowns started to occur in March 2020 [5]. However, this doesn't make the idea of using diffusion-reaction useless or invalid in all cases. Instead, it implies that the best use of such a model is to predict what would happen should the spread of the virus not be contained for the initial spread.

Our model also fails to account for individuals travelling long distances, instead assuming that everyone remains in roughly the same location. In reality, this isn't true for multiple reasons. Firstly, as a big city, people commute into central Manchester from the likes of Salford, Stockport, and Tameside both to shop and work. These people are likely to get infected in Manchester and travel back to their homes, causing the pandemic to spread to outer regions sooner than in the model. Greater Manchester isn't closed either with both national and international travel happening in and out daily. Whilst an infected person leaving Manchester wouldn't change much, at least outside of the very early days, an infected person entering Manchester could lead to a new infection centre. In reality, it turns out that, "connectivity matters more than density in the spread of COVID-19" [6].

A diffusion-reaction model also fails to account for the virus, "Sticking around". In our model, after around 30 days the area where our "patient zero" first appeared had been near completely cleared out as everyone in the area had caught the illness and recovered. In reality, not everyone in an area gets ill immediately, so the illness will stay around for longer rather than just spread to other areas. Despite that, the idea that a virus may exhaust all of its potential inhabitants in an area does have historical backing, though this is more common without lockdown protocol. PARAGRAPH NEEDS SOURCING AND NEEDS DATA.

Conclusions

In conclusion, while the diffusion-reaction equation was able to more accurately model the beginning of a disease outbreak the model starts to diverge at later times. This is primarily due to the simplifying assumptions that were made for our model. A better model would take into account the individual motion of people using a stochastic model such as a Monte-Carlo method to more accurately reflect this random walk.

References

[1] Talib Dbouk and Dimitris Drikakis.
Fluid dynamics and epidemiology: Seasonality and transmission dynamics.
Physics of Fluids, 33, 2 2021.

[2] NHS.
How to avoid catching and spreading COVID-19 infection, March 2023.

[3] Office for National Statistics.
Census-based statistics uk: 2021, 2021.

[4] Nishant Kumar, Jimi Oke, and Bat-hen Nahmias Biran.
Activity-based epidemic propagation and contact network scaling in auto-dependent metropolitan areas.
Scientific Reports, 11, 11 2021.

[5] Institute for Government analysis.
Timeline of uk coronavirus lockdowns, march 2020 to march 2021.

[6] Shima Hamidi, Reid Ewing, and Sadegh Sabouri.
Longitudinal analyses of the relationship between development density and the covid-19 morbidity and mortality rates: Early evidence from 1,165 metropolitan counties in the united states.
Health & Place, 64:102378, 2020.